

Portfolio Project: Student Course Management System using MySQL

Project Overview

A coaching institute currently maintains student and course information manually in spreadsheets. As the number of students increases, it becomes difficult to manage enrollments, update records, and generate reports.

The institute has decided to build a MySQL database to store student information, available courses, and enrollment details. As a **Data Analyst**, your responsibility is to design the database, perform CRUD operations, and answer business questions using SQL queries.

Project Objectives

- Design a relational database.
 - Perform Create, Read, Update, and Delete (CRUD) operations.
 - Retrieve information using different SQL SELECT queries.
 - Generate meaningful business reports using SQL functions and joins.
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Business Questions (Problem Statement)

You have been asked to answer the following business questions using SQL.

Section A – Database Creation

1. Create a database named **Student_Management**.
 2. Create the required tables for Students, Courses, and Enrollments.
 3. Define Primary Keys and Foreign Keys.
 4. Insert sample data into each table.
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Section B – CRUD Operations

Create

5. Add a new student named **Karan** from Hyderabad.
6. Add a new course called **Tableau** with a fee of ₹8,500.
7. Enroll Karan in the Tableau course.

Read

8. Display all student records.
9. Display all available courses.
10. Display all enrollments.

Update

11. Change Priya's city from Delhi to Bangalore.
12. Increase the SQL course fee by ₹500.
13. Update Rahul's age to 24.

Delete

14. Delete the student named Rohit.
 15. Delete the Tableau course.
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Section C – SQL SELECT Query Practice

Basic Queries

16. Display only Student Name and City.
 17. Display students whose city is Mumbai.
 18. Display female students.
 19. Display students whose age is greater than 22 years.
 20. Display students from Delhi or Pune.
 21. Display students who are NOT from Mumbai.
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Sorting

22. Display students ordered by age in ascending order.
 23. Display students ordered by age in descending order.
 24. Display courses ordered by fees from highest to lowest.
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DISTINCT

25. Display all unique cities.
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LIMIT

26. Display the first five students.
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LIKE Operator

- 27. Find students whose names start with the letter "A".
 - 28. Find students whose names end with "a".
 - 29. Find students whose names contain "it".
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BETWEEN

- 30. Find students whose age is between 21 and 23.
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IN Operator

- 31. Display students from Mumbai and Delhi.

32.

Aggregate Functions

- 32. Count the total number of students.
- 33. Find the highest course fee.
- 34. Find the lowest course fee.
- 35. Calculate the average student age.
- 36. Calculate the total fees of all courses.

GROUP BY

- 37. Count the number of students in each city.
 - 38. Count the number of students enrolled in each course.
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HAVING

- 39. Display only those cities having more than one student.
 - 40. Display courses having more than one enrollment.
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JOINS

- 41. Display each student's name along with the course they enrolled in.
 - 42. Display all students, including those who have not enrolled in any course.
 - 43. Display all courses, including those with no students enrolled.
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Alias

- 44. Display Student Name as **Name** and City as **Location**.
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Subquery

- 45. Display the names of students enrolled in the SQL course.
 - 46. Display students enrolled in the course with the highest fee.
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CASE Statement

- 47. Categorize students as:

- Senior (Age ≥ 23)
 - Junior (Age < 23)
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Date Functions

48. Display the enrollment year of each student.
 49. Display students who enrolled in the month of March.
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Bonus Analytical Questions

50. Which city has the highest number of students?
51. Which course has the highest fee?
52. Which course has the maximum enrollments?
53. Which students are enrolled in more than one course?
54. What is the average age of students in each city?
55. Which city has the youngest average age?
56. List students who have not enrolled in any course.
57. List courses that currently have no students.
58. Display the top three oldest students.
59. Display the three most expensive courses.
60. Generate a complete student report showing:

- Student Name
 - Gender
 - Age
 - City
 - Course Name
 - Course Fee
 - Enrollment Date
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Skills Covered